Spatial Data QA/QC Report

Automated Spatial Data Quality Control
Generated: 05 Oct 2026 - 16:00
Duplicate tolerance: 1.00 m

Dataset Overview

144	120	10	6	8
Total Records	Passed	Errors	Warnings	Auto Fixed

Issue Summary

Issue Type	Count	Recommended Action
Missing Coordinates	6	Manual Review
Potential Duplicates	6	Verify asset or location
X / Y Swapped	8	Automatically Corrected
Outside Valid Extent	4	Manual Review
Invalid Coordinate Values	0	Manual Review

QA/QC Method

This report was generated automatically using a PyQGIS spatial data quality-control workflow. Records were checked for missing coordinates, invalid coordinate values, locations outside the valid project extent, potential X/Y coordinate swaps, and duplicate locations. Duplicate detection was performed using a distance tolerance of 1.00 meters. X/Y values were automatically corrected only when the original coordinate pair was outside the valid project extent and the swapped coordinate pair fell inside the valid extent.

Spatial QA/QC Automation Tool
Generated using QGIS and PyQGIS

Issue Details

Records requiring manual review or automatically corrected by the QA/QC workflow.

asset_id	asset_name	category	X	Y	district	Status	Issue	Action	Duplicate With	Distance (m)	Original X	Original Y	Corrected X	Corrected Y
A0121	Utility Pole_121	Utility Pole		24.429183	Al Raha	ERROR	Missing Coordinate	Manual Review						
A0122	Sign_122	Sign	54.477018		Al Raha	ERROR	Missing Coordinate	Manual Review						
A0123	Valve_123	Valve		24.493163	Airport Area	ERROR	Missing Coordinate	Manual Review						
A0124	Manhole_124	Manhole	54.431395		Khalifa City	ERROR	Missing Coordinate	Manual Review						
A0125	Valve_125	Valve		24.386899	Masdar	ERROR	Missing Coordinate	Manual Review						
A0126	Valve_126	Valve	54.592187		Khalifa City	ERROR	Missing Coordinate	Manual Review						
A0127	Manhole_127	Manhole	54.546165	24.480265	Al Raha	WARNING	Potential Duplicate	Manual Review	A0005	0.00	54.546	24.480	54.546	24.480
A0128	Manhole_128	Manhole	54.480188	24.39399	Al Raha	WARNING	Potential Duplicate	Manual Review	A0018	0.00	54.480	24.394	54.480	24.394
A0129	Manhole_129	Manhole	54.491997	24.361282	Al Raha	WARNING	Potential Duplicate	Manual Review	A0033	0.00	54.492	24.361	54.492	24.361
A0130	Sign_130	Sign	54.493681	24.395695	Al Raha	WARNING	Potential Duplicate	Manual Review	A0047	0.00	54.494	24.396	54.494	24.396
A0131	Manhole_131	Manhole	54.442206	24.490373	Khalifa City	WARNING	Potential Duplicate	Manual Review	A0071	0.00	54.442	24.490	54.442	24.490
A0132	Manhole_132	Manhole	54.584732	24.533378	Masdar	WARNING	Potential Duplicate	Manual Review	A0096	0.00	54.585	24.533	54.585	24.533
A0133	Utility Pole_133	Utility Pole	24.425216	54.39413	Airport Area	FIXED	X/Y Swapped	Auto Corrected			24.425	54.394	54.394	24.425
A0134	Sign_134	Sign	24.393094	54.47147	Airport Area	FIXED	X/Y Swapped	Auto Corrected			24.393	54.471	54.471	24.393
A0135	Streetlight_135	Streetlight	24.453453	54.579064	Airport Area	FIXED	X/Y Swapped	Auto Corrected			24.453	54.579	54.579	24.453
A0136	Valve_136	Valve	24.510839	54.484936	Airport Area	FIXED	X/Y Swapped	Auto Corrected			24.511	54.485	54.485	24.511
A0137	Sign_137	Sign	24.512549	54.567759	Al Raha	FIXED	X/Y Swapped	Auto Corrected			24.513	54.568	54.568	24.513
A0138	Sign_138	Sign	24.40367	54.48741	Khalifa City	FIXED	X/Y Swapped	Auto Corrected			24.404	54.487	54.487	24.404
A0139	Sign_139	Sign	24.46702	54.465549	Al Raha	FIXED	X/Y Swapped	Auto Corrected			24.467	54.466	54.466	24.467
A0140	Utility Pole_140	Utility Pole	24.507127	54.564911	Khalifa City	FIXED	X/Y Swapped	Auto Corrected			24.507	54.565	54.565	24.507
A0141	Sign_141	Sign	55.2	24.45	Unknown	ERROR	Outside Valid Extent	Manual Review			55.200	24.450		
A0142	Utility Pole_142	Utility Pole	54.45	25.1	Unknown	ERROR	Outside Valid Extent	Manual Review			54.450	25.100		
A0143	Utility Pole_143	Utility Pole	53.9	24.4	Unknown	ERROR	Outside Valid Extent	Manual Review			53.900	24.400		
A0144	Streetlight_144	Streetlight	54.5	23.95	Unknown	ERROR	Outside Valid Extent	Manual Review			54.500	23.950		